TAGUIG CITY UNIVERSITY COLLEGE OF INFORMATION AND COMPUTING TECHNOLOGY

A Risk-Based Two-Factor Student Verification and Event Attendance Management System

Using Near Field Communication (NFC), Personal Identification Number (PIN) Authentication, and Quick Response (QR) Credential Validation

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Pre-oral Recommendations on Evidence

Recommendation	Action and Present Status
01-02 Checklist and Chapter 4 visuals	Complete checklist, corrected results, item tables, and comparison figures.
03 Raw survey and comparison records	Student workbook and exports available. Full anonymized records still need a traceable evidence package.
04 Evidence for tests and surveys	16 listed test cases and appendix photographs. Evidence does not cover every current feature.
05 Interfaces in appendices	Appendix D includes selected interfaces. Current timestamp and maintenance screenshots remain to be added.

Selected evidence appears in Appendices D and H-K. Pending evidence remains identified.

Pre-oral Recommendations on Comparison

Recommendation	Action and Present Status
06-07 Manual questionnaire and data	Manual questionnaire, 70 writing observations, and 70 manual survey responses included.
08 Student participant counts	70 manual students and 70 different system students. System survey also includes two groups of five.
09-10 Same students and survey sequence	Not fulfilled as specified. Different groups require an independent-samples analysis.
16 Use case diagram	Diagram included. Conditional PIN use still needs alignment with NFC-only Fast mode.
17 T-test of manual and system means	Welch test compares overall survey scores. Equivalent complete-task times were unavailable.

Equal group sizes do not mean the same participants or eliminate group-selection bias.

Pre-oral Recommendations on the System

Recommendation	Current Implementation and Qualification
11 Personnel login timestamp	Timestamped login audit events exist. A dedicated visible login-time field remains unverified.
12 Student timestamps	Kiosk code displays scan time and applicable IN/OUT values.
13 Retain student result	Configured 10-second result display. Persistent last-log display after idle remains a separate request.
14 Administrator purge	Authorized, confirmed active cleanup of eligible cloud-backed local logs.
15 Data destruction	Partial. Local cleanup retains cloud history. Complete disposal across copies is not established.
18 Spell out NFC in the title	Latest title spells out Near Field Communication and the PIN/QR terms.

Current code confirms implementation details. Dated UI and deletion tests are still needed.

Problem and Research Objectives

Problem

Campus verification and event attendance require credential checks, eligibility decisions, and traceable records. Paper sign-in alone does not perform these checks.

Development objective

Develop a prototype that combines configurable credential validation with entry/exit handling, event attendance, and role-based monitoring.

Evaluation objective

Describe system metrics and perceived quality. Compare manual and system survey ratings while preserving the limits of the timing data.

Scope and Configured Verification Modes

Mode	Required Credentials	Selection
Fast	NFC only	Operator or event configuration
Standard	NFC and PIN	Operator or event configuration
High-Security	NFC, PIN, and QR validation	Operator or event configuration

QR checks consistency with the NFC-linked credential record. It does not independently prove physical identity.

The evaluated prototype used configured modes rather than a per-student automatic risk score.

Research Design and Respondents

Quantitative developmental and descriptive-evaluative design

Iterative Waterfall development with controlled observation and ISO/IEC 25010-based questionnaires. Purposive and convenience sampling.

Survey Group	n	Evaluation
Students, proposed system	70	Five common quality categories
Students, manual process	70	Corresponding five-category questionnaire
Security personnel	5	Five quality categories
Faculty, event organizer, administrators	5	Eight quality categories

150 survey respondents in total. The two student survey groups contain different people.

Controlled Observation Procedure

Session	Participants	Entry Rounds per Mode
August 12, 2026	40	20 entries, reset exits, then 20 entries
August 13, 2026	30	20 entries, reset exits, then 10 entries

Same participants across modes

70 participants completed Fast, Standard, then High-Security in that fixed order. Each mode contains 70 successful entries.

Shared test credentials

20 preconfigured NFC cards with PIN and QR were reused. Team exit transactions reset the profiles between rounds.

System Architecture and Evaluation Scope

Layer	Role
ESP32 and PN532	Read NFC UID and transmit it to the Windows application through USB serial.
WinUI 3 application	Apply configured rules, validate PIN/QR, assign timestamps, and update displays and records.
Local MySQL	Store operational student, event, attendance, and security records.
Firebase Firestore	Support cloud synchronization and backup.

Added after the August observation

Signed QR credentials, durable offline queues, event-completion rules, and graduation clearance require separate evaluation evidence.

Statistical Treatment and Hypotheses

Descriptive analysis

Frequency, mean, and sample SD use applicable observations. Respondent category scores produce the category and overall composite SDs.

Welch independent-samples t-test

Two-tailed test of overall student survey scores at alpha = 0.05. Different students form the manual and proposed-system groups.

H0: No significant difference in mean overall survey ratings.

H1: A significant difference exists in mean overall survey ratings.

t = (mean System - mean Manual) / sqrt(SD System^2 / n System + SD Manual^2 / n Manual)

Observation Records and Functionality Cases

Mode	Successful Entries	Reset Exits	Denied Records	Total
Fast	70	70	2	142
Standard	70	70	12	152
High-Security	70	70	4	144
Total	210	210	18	438

Recorded challenges

Denials include deliberate PIN/lockout, QR mismatch, and entry/exit state tests.

Documented functionality

Appendix H lists 16 PASS cases. These do not validate every feature or later addition.

Internal System Latency

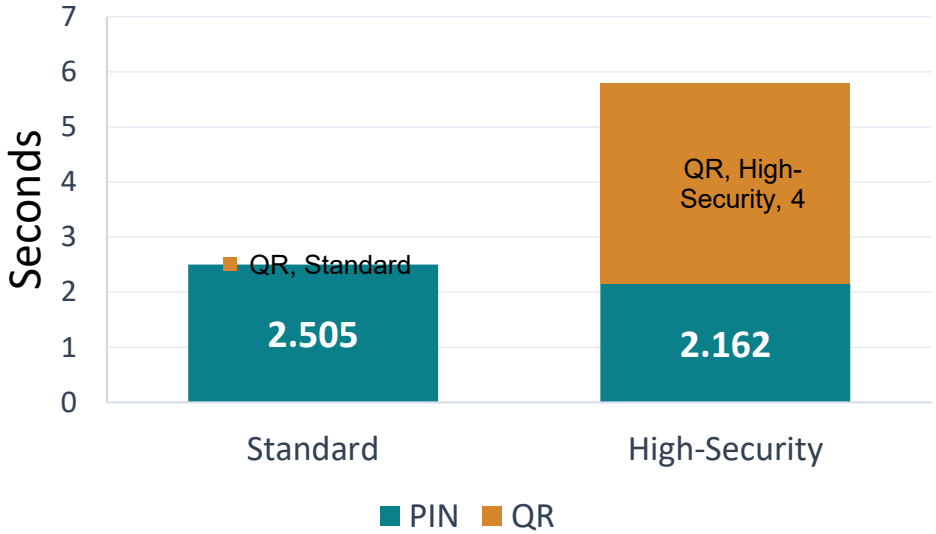
Mode	Mean ms	Sample SD ms	DB Mean ms
Fast	11.49	2.20	2.88
Standard	57.37	17.46	47.95
High-Security	54.06	6.34	44.90



70 successful entries per mode

Database-query time overlaps internal latency. Adding it again would double-count time.

Recorded PIN and QR Interaction



Total Stage Duration	Standard	High-Security
Mean (s)	2.505	5.799
Median (s)	2.035	4.348
Sample SD (s)	1.152	3.878
Maximum (s)	6.875	22.604

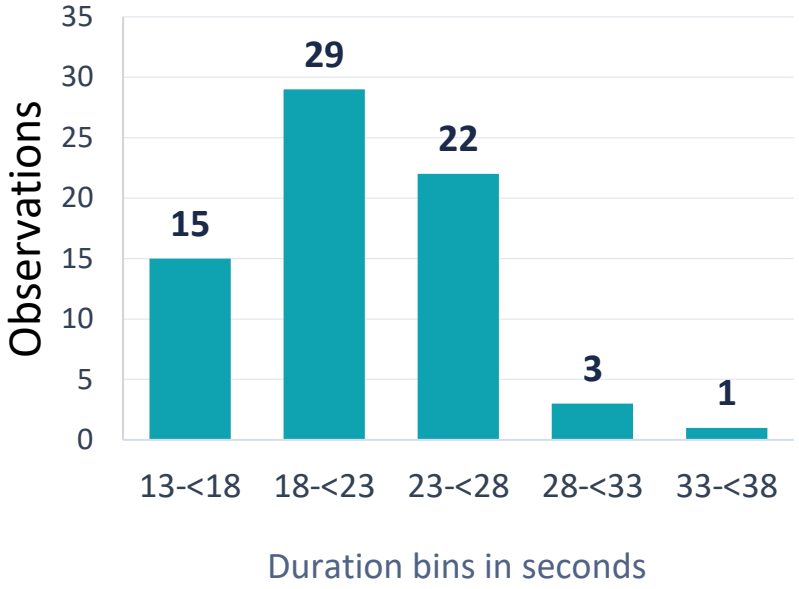
n = 70 per mode. Fast bypasses PIN/QR. Its interaction stage is N/A, not a zero-second full task.

Manual Paper Attendance Baseline

21.803 seconds

Mean active writing duration, n = 70

Sample SD 4.436 s
Median 21.135 s
Range 13.21-37.63 s



Stopwatch: first written character to completed name, course/section, student ID, and signature.

Recorded Entry Spacing and Rate

Mode	Intervals	Mean Spacing s	Recorded Rate / min
Fast	66	2.730	21.98
Standard	66	8.048	7.46
High-Security	66	12.129	4.95

Why 66 intervals?

Rounds of 20, 20, 20, and 10 entries produce 19 + 19 + 19 + 9 gaps between consecutive entries.

These rates do not establish full-round capacity or a matched throughput advantage over manual attendance.

Credential Decisions and Security Conditions

100.00%

222 / 222 correct applicable credential decisions

4.11%

18 / 438 denied exported records

1.32%

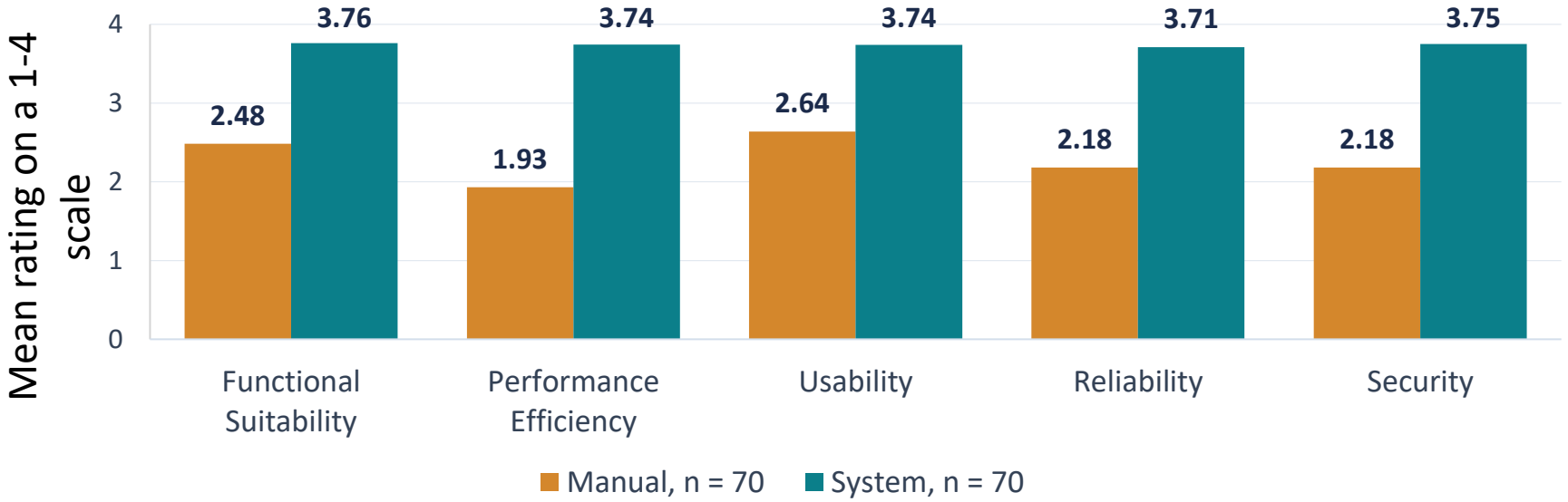
2 / 152 newly triggered PIN lockouts

Recorded Denial Condition	Count
Incorrect PIN	8
Consecutive-entry state violation	5
New PIN lockout	2
QR mismatch	2
Irregular exit state	1

Expected outcomes came from the researcher. Demonstrations contribute to denial and lockout counts.

Controlled credential decisions do not establish physical identity or universal attack resistance.

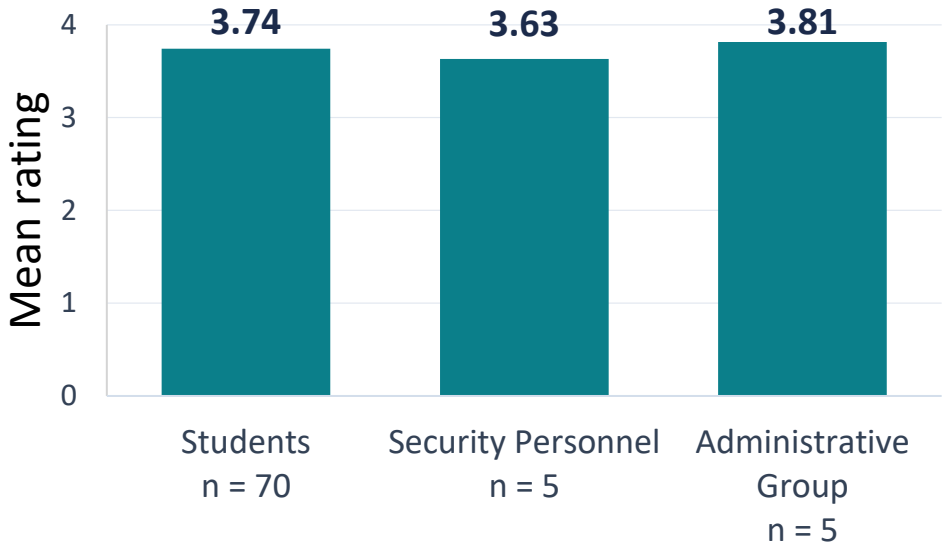
Manual and System Student Survey Ratings



Largest descriptive difference: Performance Efficiency, 1.811 points.

Different student groups. Category means are descriptive. Only the overall score received a t-test.

Proposed-System Evaluation by Group



Highly Satisfactory

All three overall means fall in the 3.26-4.00 interpretation range.

Security personnel: five categories
Lowest category mean: Performance Efficiency, 3.45.

Administrative group: eight categories
Portability 3.92, Performance Efficiency 3.68.

Role-specific instruments remain separate. No pooled system-wide mean or composite SD for the five-person groups.

Welch Test of Overall Survey Scores

Group	n	Overall Mean	Composite Sample SD
Manual process	70	2.284	0.753
Proposed system	70	3.742	0.290

Rounded demonstration

SE = sqrt(0.290^2 / 70 + 0.753^2 / 70) = 0.096445
t = (3.742 - 2.284) / 0.096445 = 15.12

t(89.076) = 15.124, p < .001

Difference 1.458 points
95% CI [1.267, 1.650]

Reject H0 at alpha = .05. The result concerns overall survey ratings, not processing speed.

Discussion and Study Limitations

Recorded capability

The prototype applied the configured credential and state rules in the recorded controlled tests. Logical re-entry rejection does not prove physical tailgating prevention.

Timing interpretation

Millisecond internal latency and seconds of PIN/QR interaction describe different components. Complete-task speed and matched throughput remain unmeasured.

Evaluation limits

Shared cards, fixed mode order, non-probability sampling, small staff groups, and different student survey groups limit generalization and causal interpretation.

Conclusion

Developed prototype

The system combines configured NFC/PIN/QR workflows, eligibility checks, attendance records, and role-based monitoring.

Supported findings

Recorded credential decisions matched expected outcomes in the controlled tests. Respondents reported favorable perceived quality, with higher overall student ratings for the system group.

Scope of the conclusion

The study supports demonstrated prototype functionality and the survey comparison. Deployment-scale performance and broader security effectiveness require further testing.

Recommendations for Further Evaluation

Equivalent timing and larger studies

Measure full tasks and sustained throughput with common boundaries. Use matched or clearly independent groups and counterbalance mode order.

Security, recovery, and personnel workflows

Test credential sharing and copied payloads, outages and power recovery, alert workload, and routine PIN usability separately from demonstrations.

Documentation and deployment readiness

Retain anonymized raw responses, complete dated evidence, and current interface screenshots. Resolve retention/destruction scope before wider deployment.

SYSTEM DEMONSTRATION

Live Demo and Questions

Configured modes, entry/exit timestamps, alerts, and authorized local cleanup